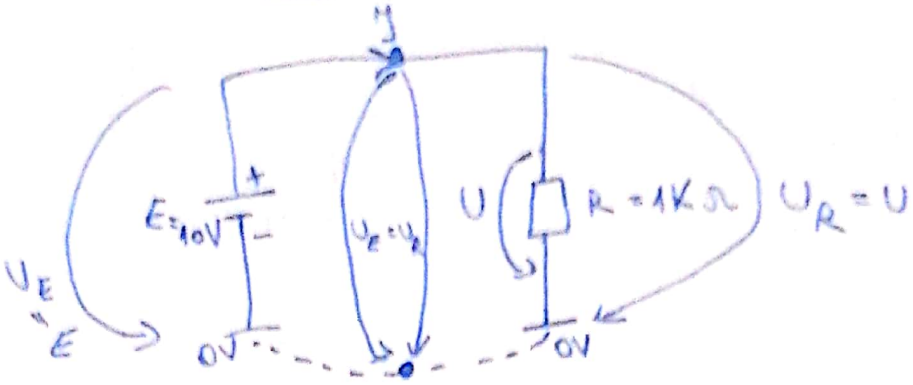
13.X.2022

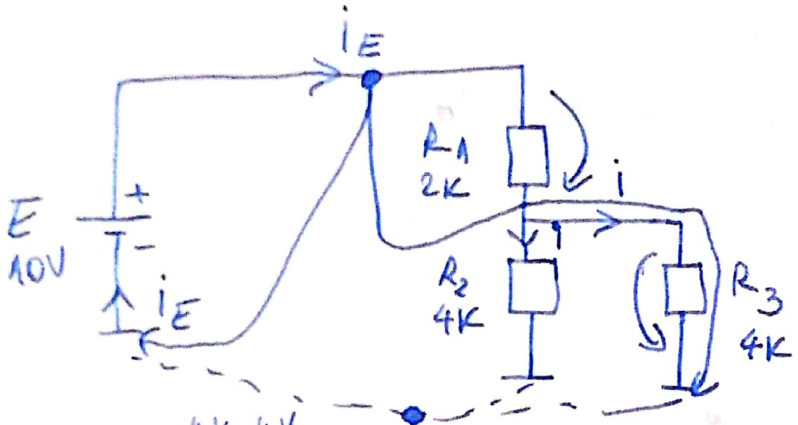
Seminar DEEA I

- OHM
- KCL KVL
- THEVENIN



$U = Ri$

$E = Ri \Rightarrow i = \frac{E}{R} = \frac{10V}{1K\Omega} = 10mA$

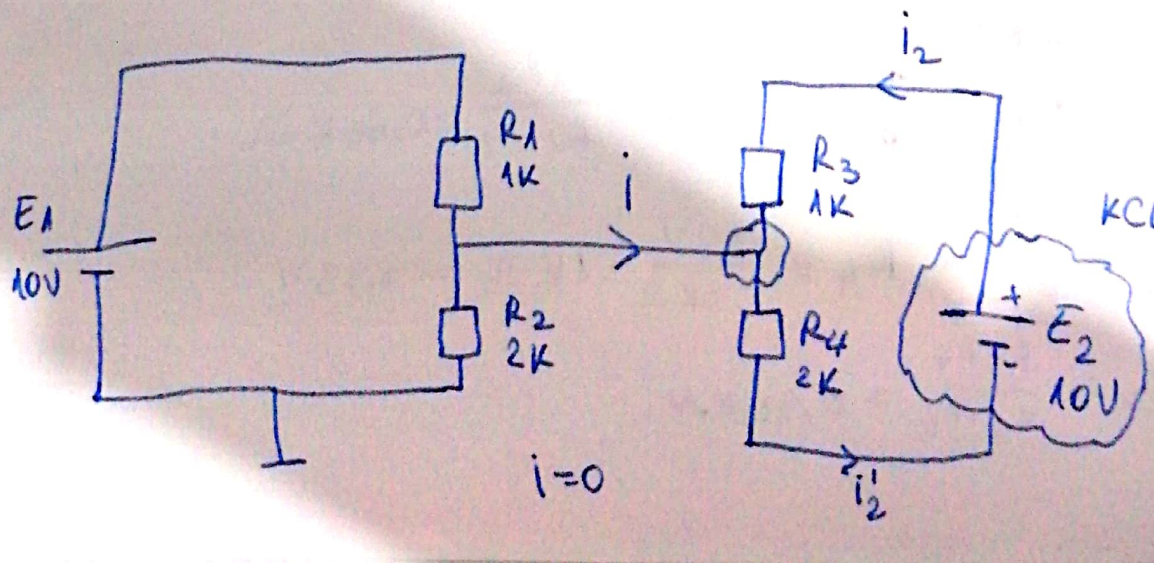


$i = ?$

$R_e = \frac{4K \cdot 4K}{8K} + 2K = 4K$

$i_E = \frac{10V}{4K} = 2,5\,mA$

$i = \frac{i_E}{2} = 1,25\,mA$



$i = ?$

KCL: $i_2 - i_2' = 0$

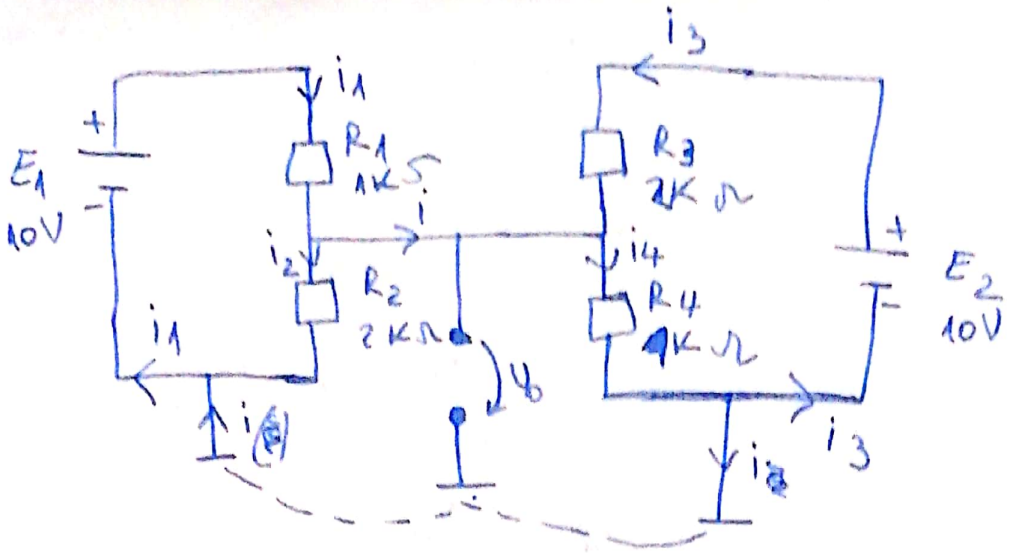
$i_2 = i_2'$

$i = 0$

$i + i_2 - i_2' = 0$

$i = 0$

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$i = ?$

$U_o = ?$

$i_1 = i + i_2$

$i_4 = i + i_3$

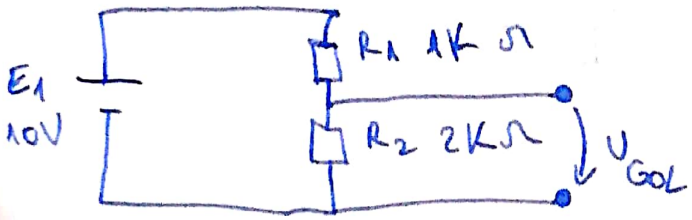
U_o

$E_2 = i_3 R_3 + (i_4 R_4 + E_2 = \dots)$

$E_1 = i_1 R_1 + i_2 R_2$, $E_1 = i_1 R_1 + U_o$

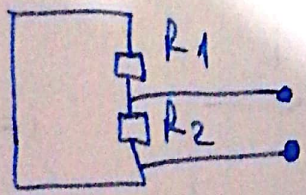
$U_o = R_2 i_2$

$U_o = R_4 i_4$



$E_1 = i_1 (R_1 + R_2) \Rightarrow i_1 = 3{,}33\ mA$

$U_{GOL} = i_1 R_2 = 6{,}66\ V = E_{e1}$

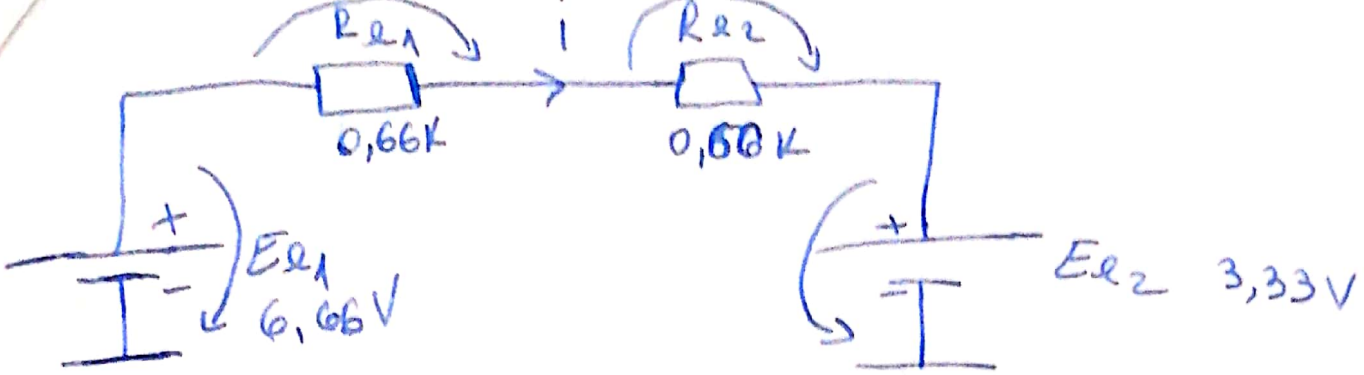


$R_{e1} = \frac{R_1 R_2}{R_1 + R_2} = 0{,}66\ K\Omega$

$E_{e2} = \frac{E_2}{R_3 + R_4} \cdot R_4 = \frac{10\ V}{3K\Omega} \cdot 1K\Omega = 3{,}33\ V$

$R_{e2} = \frac{R_3 R_4}{R_3 + R_4} = 0{,}33\ K\Omega$

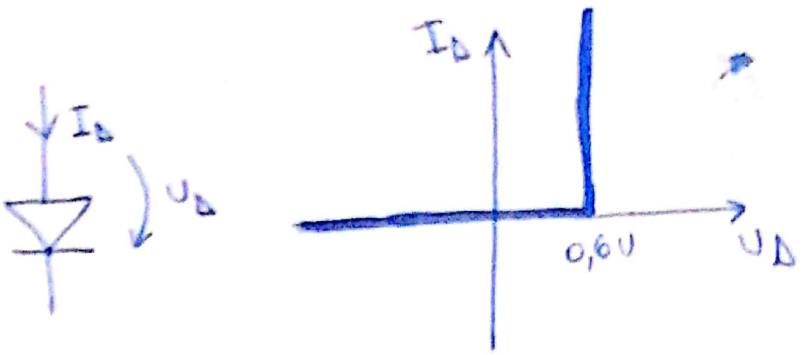
Scanned with CamScanner



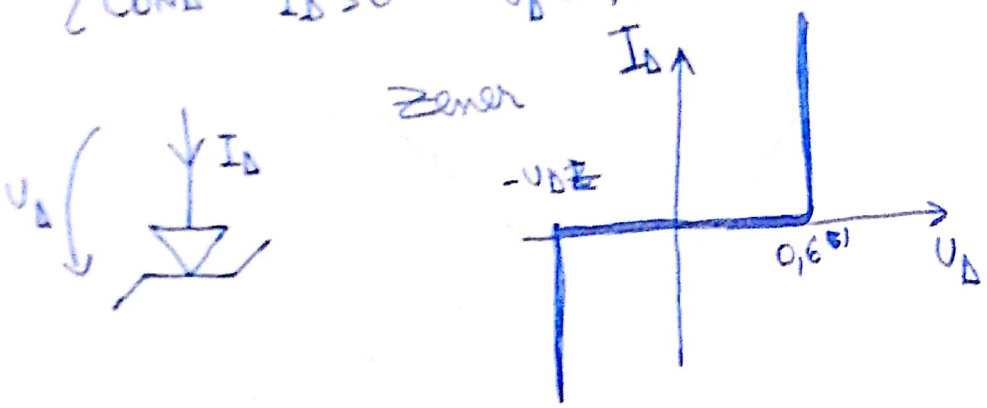
$$E_{\ell 1} - E_{\ell 2} = iR_{\ell 1} + iR_{\ell 2} \Rightarrow i = \frac{E_{\ell 1} - E_{\ell 2}}{R_{\ell 1} + R_{\ell 2}} = \frac{3{,}33\,V}{1{,}32\,K\Omega} =$$

$$= 2{,}522\,mA$$

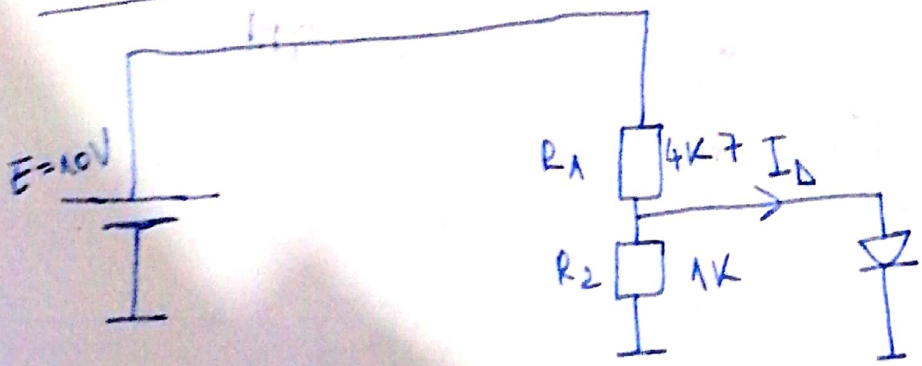
Scanned with CamScanner



BL	$U_D < 0,6V$	$I_D = 0$
COND	$I_D > 0$	$U_D = 0,6V$

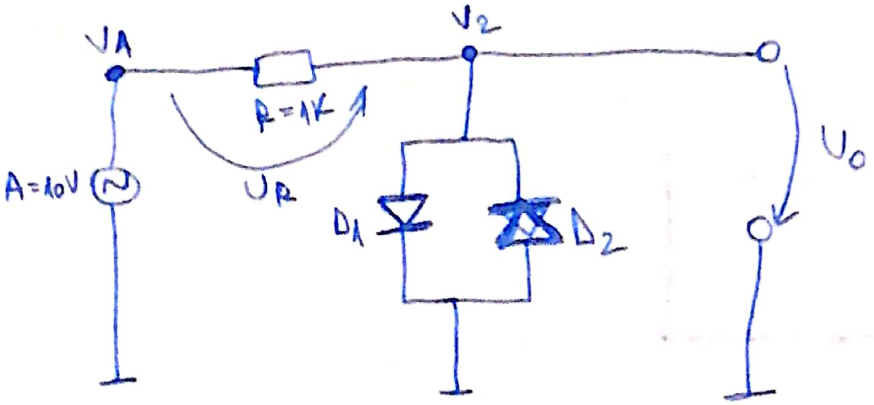


COND. D.	$I_D > 0$	$U_D = 0,6$
BL	$I_D = 0$	$-U_{Dz} < U_D < 0,6$
COND. INV.	$I_D < 0$	$U_D = -U_{Dz}$

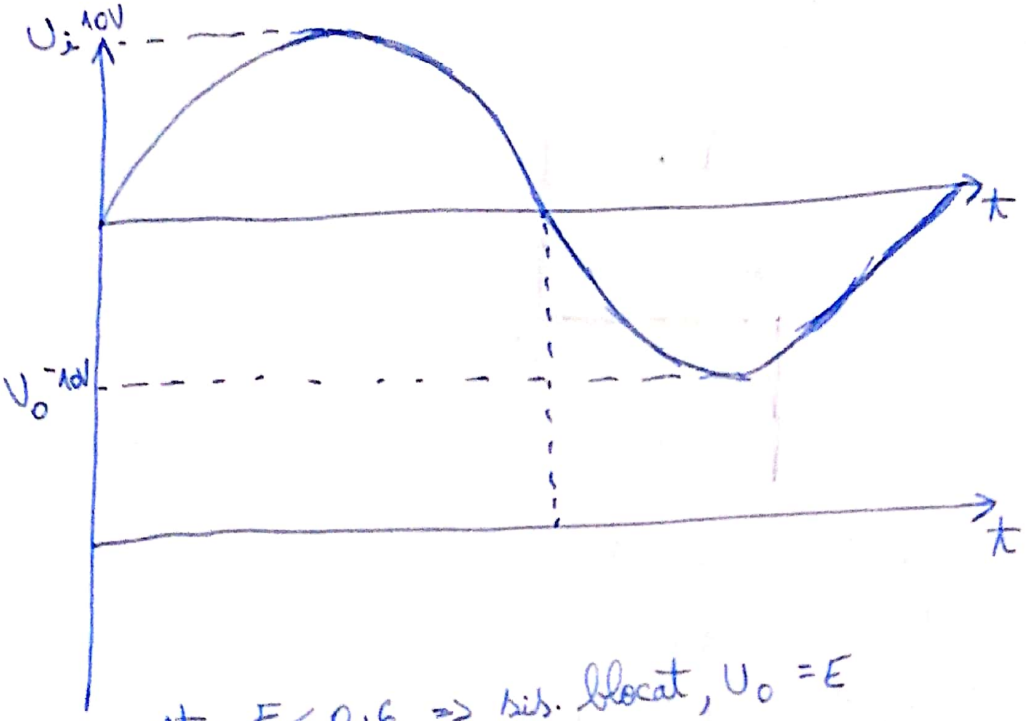


Se presupune $I_D > 0$, $U_D = 0,6V$

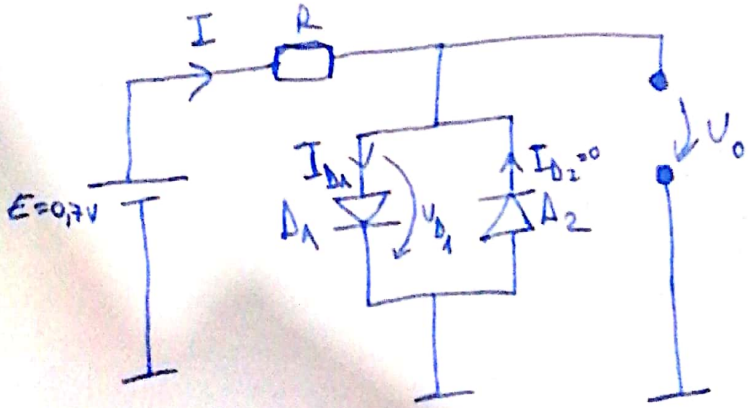
Scanned with CamScanner



$U_R = V_1 - V_2$



pt. $E < 0,6$ => sis. blocat, $U_0 = E$



Pp. că D_1 = CONL, D_2 = BL

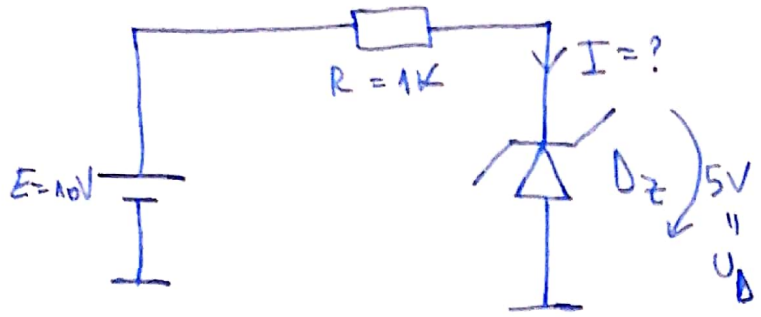
$U_{D_1} = 0,6V$ ⇓ $I_{D_2} = 0A$

$U_0 = U_{D_1} = 0,6V$

$I_{D_1} = I$

$I = \frac{E - U_0}{R} > 0$

Scanned with CamScanner

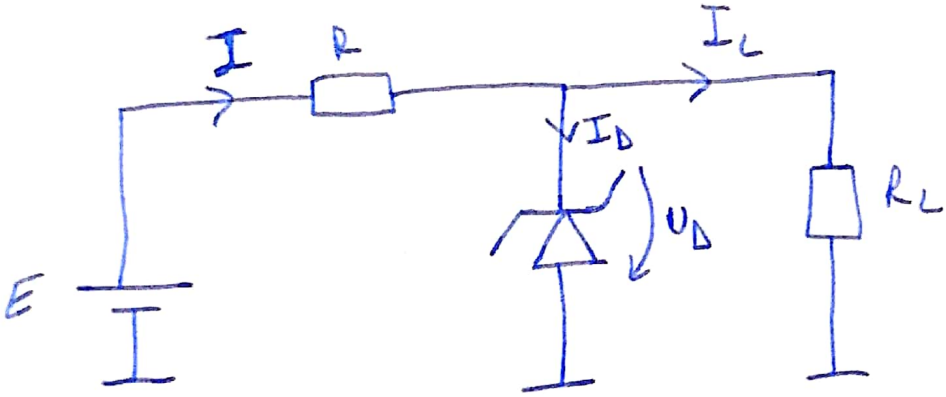


$U_{Dz} = 5V$

$I = ?$

Pt. că D_z e în COND. INV. $\Rightarrow U_D = 5V$

$E = IR + U_D \Rightarrow I = \frac{E - U_D}{R} = 5mA > 0 \Rightarrow$ Pt. e adev.



$E = 10V$

$U_{Dz} = 5V$

$R_L = 100\Omega$

$10V < E < 12,6$

$R_{MAX} = ?$ a.î. $I_{D\,min} = 5mA$

$I = I_D + I_L$, $I_L = \frac{U_D}{R_L} = \frac{5V}{100\Omega} = 50mA$

$U_D = 5V$

$E = IR + U_D = R(I_D + I_L) + U_D$

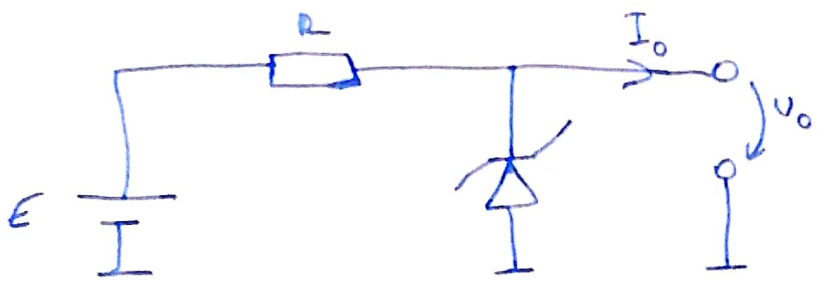
$\Rightarrow R = \frac{E - U_D}{I_D + I_L} = \frac{5V}{(5mA) + 50mA} = \frac{5V}{55mA} \cong 90\Omega$

$I_D = \frac{E - U_D - I_L R}{R} \geq 5mA$, (H) E

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Seminar 4

8 · XII · 2022



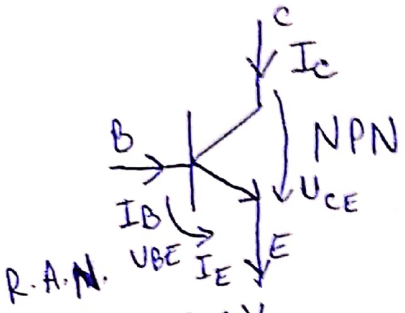
$$\begin{cases} 10\,V < E < 12\,V \\ 0 < I_0 < 100\,mA \\ I_\Delta \geq 5\,mA \end{cases}$$

$R_{MAX} = ?$

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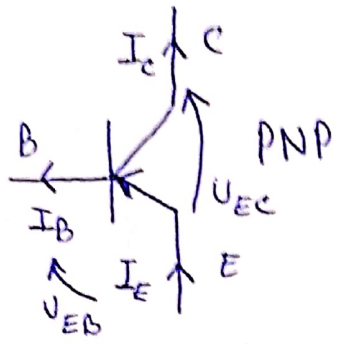
Seminar 3
AEEA

P.S.F.

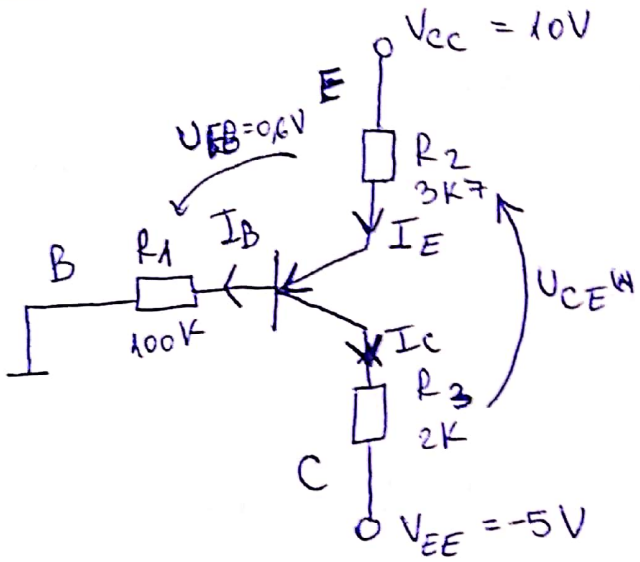


R.A.N.
$U_{BE} = 0,6\,V$
$\Rightarrow U_{CE} > U_{BE}$

$I_E = I_C + I_B$
$I_C = \beta I_B$



$U_{EB} \cong 0,6\,V$
$U_{EC} > U_{EB}$



$\beta = 100$
$I_C = ?$, $U_{CE} = ?$

Ip. RAN $\Rightarrow$ $U_{EB} = 0,6\,V$
$I_C = \beta I_B$

$V_{CC} - 0 = I_E R_2 + U_{EB} + I_B R_1$

$(I_B + \beta I_B) \approx I_C$

$V_{CC} - 0 = I_C R_2 + U_{EB} + \frac{I_C}{\beta} R_1$

$10V = I_C \cdot 3,7K + 0,6V + \frac{I_C}{100} \cdot 100K$

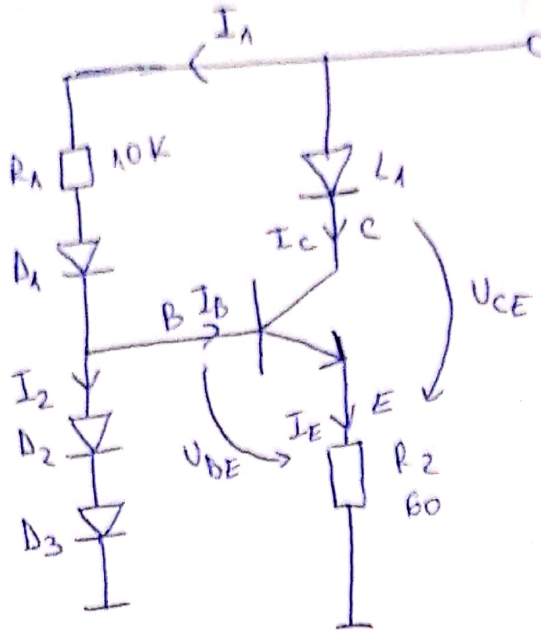
$\frac{9,4\,V \cdot 100}{470K} = I_C = 2\,mA$

$V_{CC} - V_{EE} = I_E R_2 + U_{CE} + I_C R_3$

$U_{CE} = V_{EE} + I_C(R_2 + R_3) - V_{CC} =$

$= -5V + 2\,mA \cdot 5,7K - 10V = -3,6\,V$

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$U_{D_{1,2,3}} = 0,6\,V$ în cond.

$U_{L_1} = 3\,V$ în cond.

$\beta = 100$

$I_C = ?$ $U_{CE} = ?$

Pp. că bunt în cond.

$$U_{D_2} + U_{D_3} = U_{BE} + I_E R_2$$

$$\Rightarrow I_E \approx \frac{2U_D - U_{BE}}{R_2} = 10\,mA$$

$$V_{CC} - 0 = U_{L_1} + U_{CE} + I_E R_2$$

$$U_{CE} = V_{CC} - U_{L_1} - I_E R_2 = 10V - 3V - 0,6V = 6,4\,V > 0,6$$

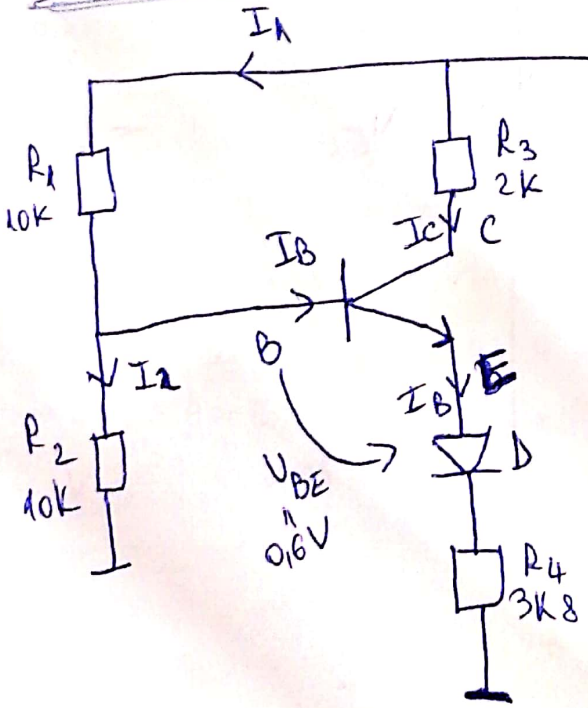
$\Rightarrow$ RAN

$$V_{CC} - 0 = I_1 R_1 + U_{D_1} + U_{D_2} + U_{D_3}$$

$$\Rightarrow I_1 = \frac{V_{CC} - 3U_D}{R_1} = \frac{8,2\,V}{10K} = 0,82\,mA$$

$$I_1 = I_2 + I_B \Rightarrow \cancel{I_B = I_1 - I_2} \quad I_2 = I_1 - I_B = I_1 - \frac{I_C}{\beta} =$$

$= 0,72\,mA$



$\beta = 100$

PSE: I_C, V_{CE}

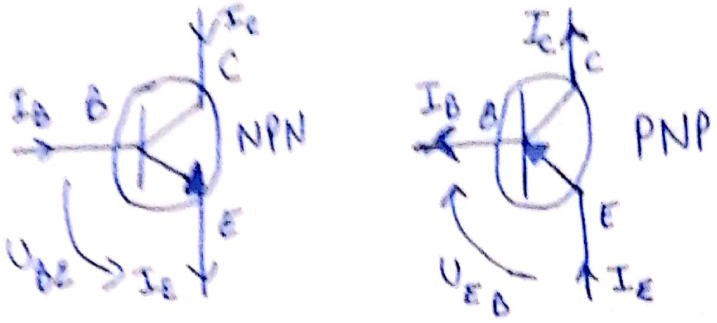
Pp. că D în cond. $\Rightarrow U_D = 0,6\,V$

$$\begin{cases} V_{CC} = I_1 R_1 + U_{BE} + U_D + I_E R_4 \\ I_2 R_2 = U_{BE} + U_D + I_E R_4 \\ I_1 = I_B + I_2 \end{cases}$$

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Seminar 4

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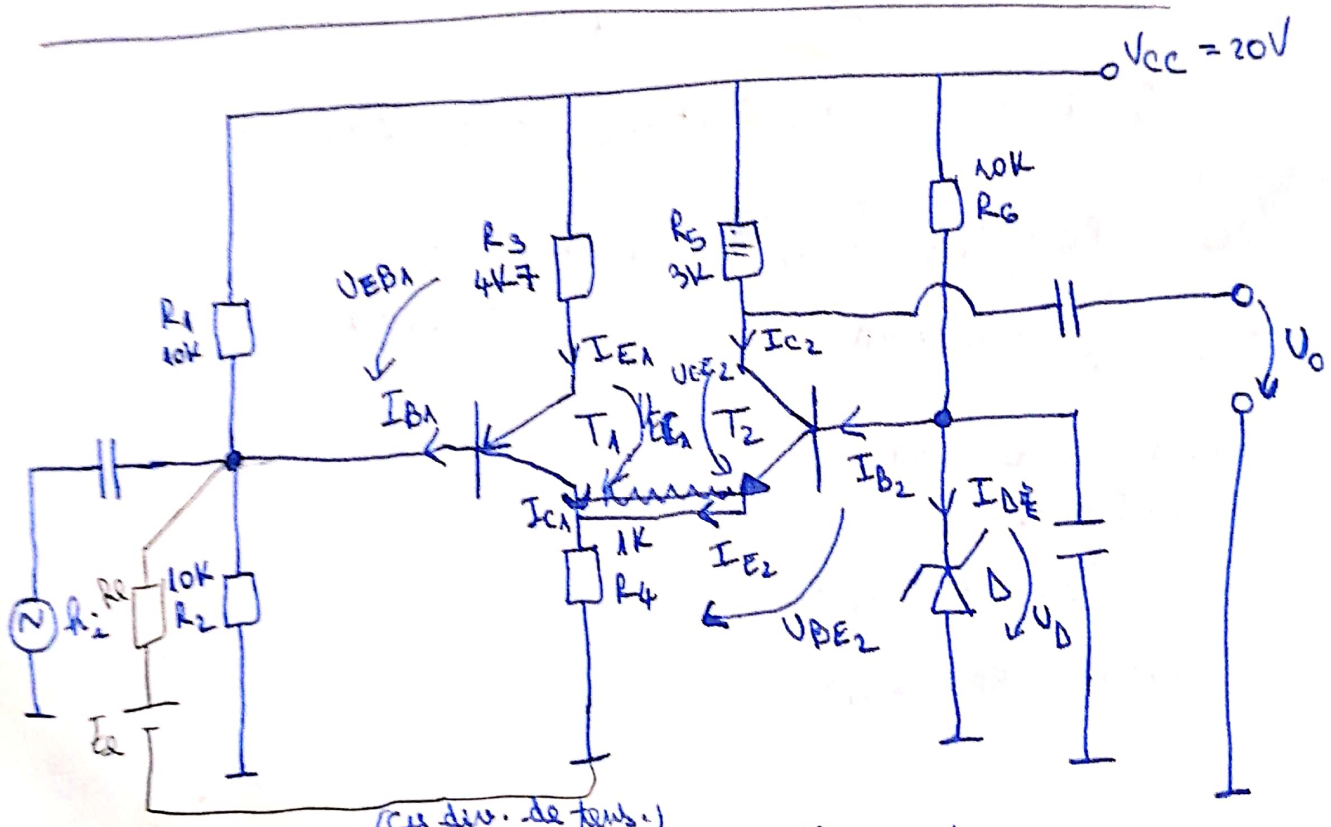


RAN: $I_C = \beta I_B$

$U_{BE} = 0,6$

$I_B + I_C = I_E \quad I_E \cong I_C$

$U_{CE} > 0,6 \quad U_{EC} > 0,6$



$U_{DZ} = 3V6$

P.S.F. I_C

U_{CE}

① (Cu div. de tens.)

PP. $I_D > 0 \Rightarrow U_D = U_{DZ} = 3V6$

$R_2 = R_1 \| R_2 = 5K$

$E_2 = \frac{(V_{CC} - 0) R_2}{R_1 + R_2} = \frac{20 \cdot 10}{20} = 10V$

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$V_{CC} = I_{E_1} R_3 + U_{EB_1} + I_{B_1} R_2 + E_2 =$

$= I_{C_1} R_3 + U_{EB_1} + \frac{I_{C_1}}{\beta} R_2 + E_2$

$\Rightarrow I_{C_1} = \frac{V_{CC} - U_{EB_1} - E_2}{R_3 + \frac{R_2}{\beta}} \cong 2\,mA$

$U_D = U_{BE_2} + R_4 (I_{C_1} + I_{E_2})$

$I_{E_2} = I_{C_2}$

$\Rightarrow I_{C_2} = \frac{U_D - U_{BE_2}}{R_4} - I_{C_1} = \frac{3,6V - 0,6V}{1K} - 2\,mA \cong 1\,mA$

$V_{CC} = I_{E_1} R_3 + U_{EC_1} + R_4 (I_{C_1} + I_{E_2})$

$\Rightarrow U_{EC_1} = V_{CC} - I_{C_1} R_3 - R_4 (I_{C_1} + I_{C_2}) \cong 7,6V > 0,6V$

$V_{CC} = I_{C_2} R_5 + U_{CE_2} + R_4 (I_{C_1} + I_{E_2}) \Rightarrow U_{CE_2} \cong 14V > 0,6V$

$I_D > 0$

$V_{CC} = I_6 R_6 + U_{DZ} \Rightarrow I_6 \cong 1,64\,mA$

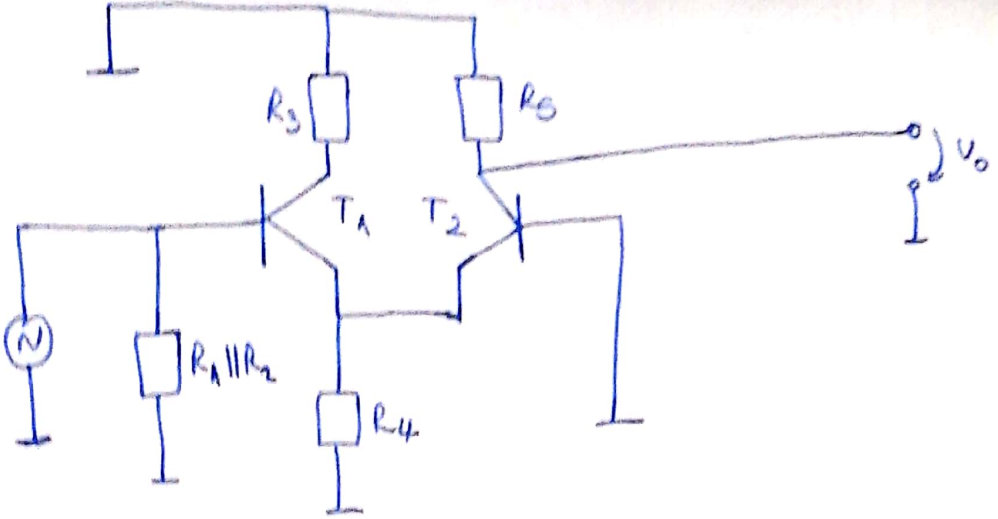
$I_{B_2} = \frac{I_{C_2}}{\beta} = 0,01\,mA$

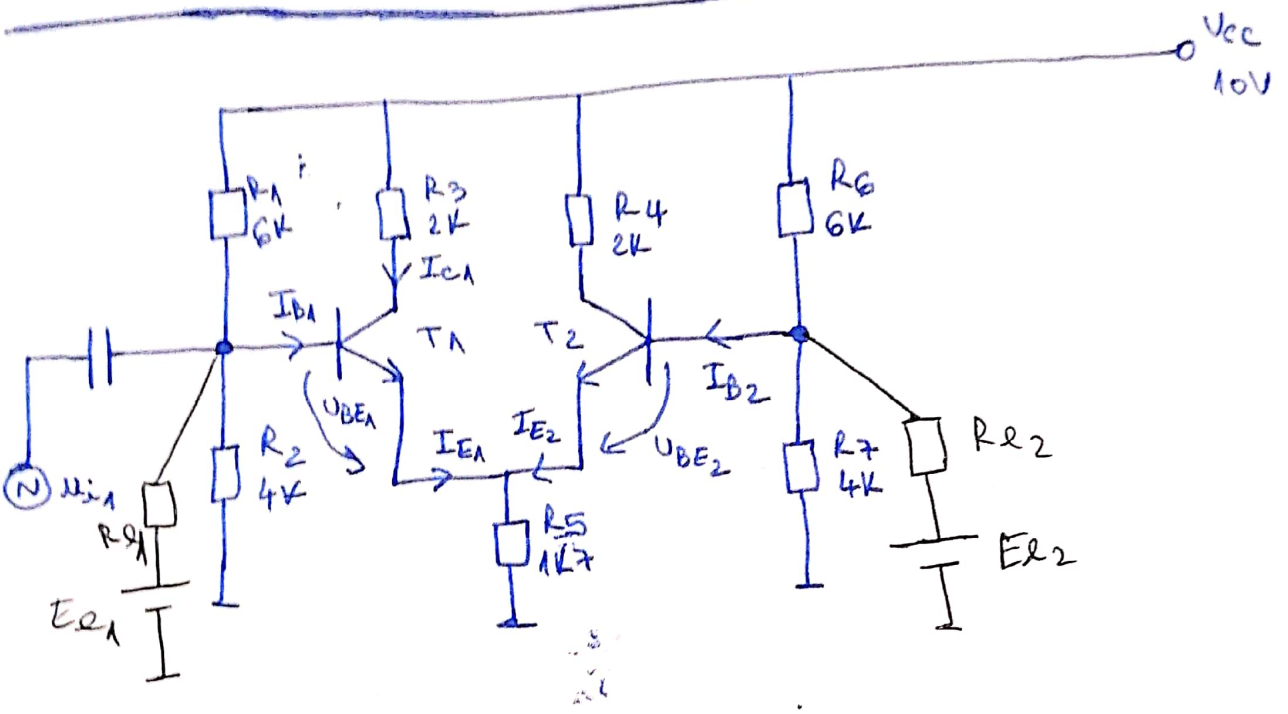
$\left.\begin{array}{l} I_D = I_6 - I_{B_2} \\ \Rightarrow I_D = 1,63\,mA > 0 \end{array}\right\}$

(Fără div. de tens.)

② $\begin{cases} I_1 + I_{B_1} = I_2 \\ V_{CC} = I_{E_1} R_3 + U_{EB_1} + I_2 R_2 \\ I_1 R_1 = U_{EB_1} + I_{E_1} R_3 \end{cases}$

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P.S.F. I_C ; U_{CE}

$R_{s_1} = R_1 \| R_2 = 2K4 = R_{s_2}$

$E_{s_1} = \frac{V_{CC} \cdot R_2}{R_1 + R_2} = 4V = E_{s_2}$

Schema e sim. $\Rightarrow I_{C_1} = I_{C_2}$

$U_{CE_1} = U_{CE_2}$

$E_1 = I_{B_1} R_{s_1} + U_{BE_1} + 2 I_{E_1} R_5$

$I_{B_1} = \frac{I_{C_1}}{\beta}$; $I_{E_1} = I_{C_1}$

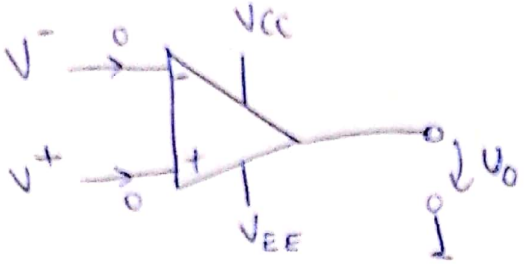
... (Am ieșit la tablă și am fost f. bun)

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Seminar 5
DEEA

12.I 2023

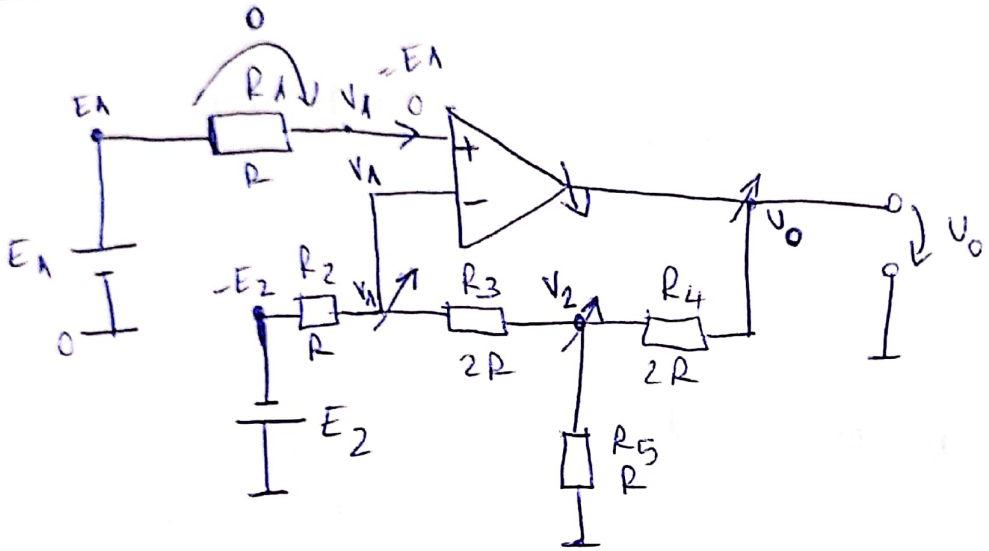
Tranzistor



$U_0 = A(V^+ - V^-)$

$A \to \infty \quad 10^6$

R.N. $\Rightarrow V^- = V^+$



E_1, E_2

$U_0 = ?$

Avem R.N. $\Rightarrow V^+ = V^-$

$$\frac{-E_2 - E_1}{R_2} + \frac{V_2 - E_1}{R_3} = 0$$

$$\frac{-E_2 - E_1}{R} + \frac{V_2 - E_1}{2R} = 0 \mid \cdot 2R$$

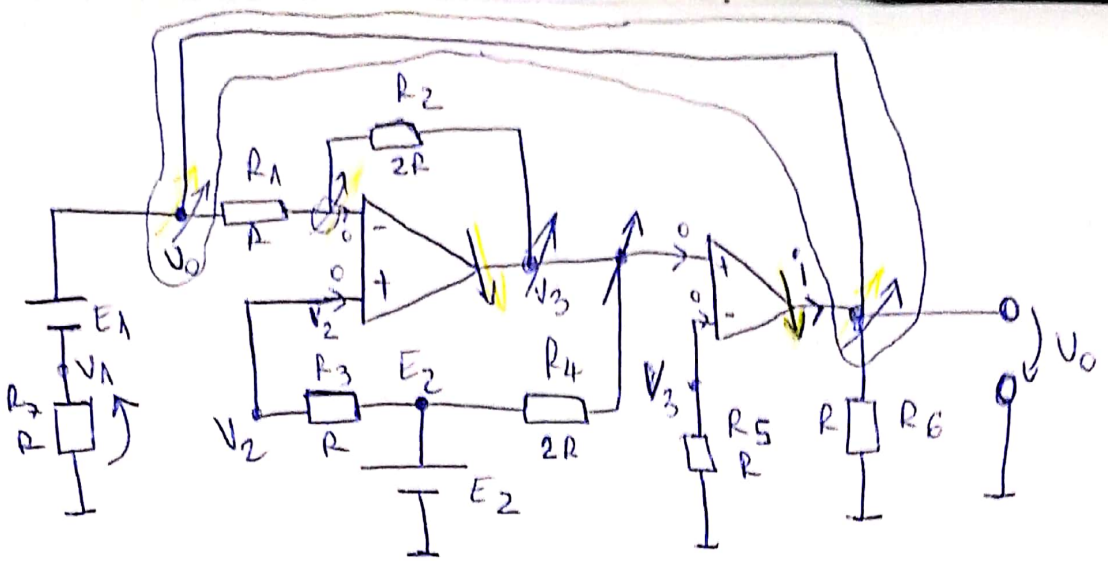
$$-2E_2 - 3E_1 + V_2 = 0 \Rightarrow V_2 = 3E_1 + 2E_2$$

$$\frac{V_1 - V_2}{R_3} + \frac{0 - V_2}{R_5} + \frac{U_0 - V_2}{R_4} = 0$$

$$\frac{V_1 - V_2}{2R} + \frac{-V_2}{R} + \frac{U_0 - V_2}{2R} = 0 \mid \cdot 2R$$

$$V_1 - V_2 - 2V_2 + U_0 - V_2 = 0 \Rightarrow U_0 = 4V_2 - V_1 = 11E_1 + 8E_2$$

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$V_0 - V_1 = E_1$

$V_2 = E_2$

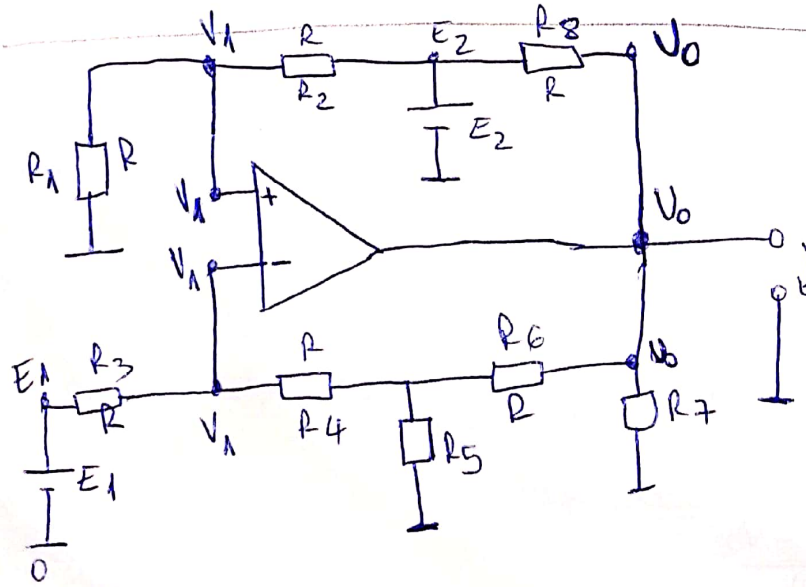
$V_3 = 0$

$$\frac{0 - V_1}{R_7} + \frac{V_2 - V_0}{R_1} + \frac{0 - V_0}{R_6} + i = 0 \quad \text{(AŞA NU)}$$

$$\frac{V_0 - V_2}{R_1} + \frac{V_3 - V_2}{R_2} = 0$$

$$2(V_0 - E_2) - E_2 = 0 \Rightarrow 2V_0 = 3E_2 \Rightarrow V_0 = \frac{3}{2}E_2$$

$$\Rightarrow V_1 = \frac{3}{2}E_2 - E_1$$



$$\frac{0 - V_1}{R_1} + \frac{E_2 - V_1}{R_2} + 0 = 0$$

$$\Leftrightarrow \frac{-V_1}{R} + \frac{E_2 - V_1}{R} = 0$$

$$\Leftrightarrow E_2 = 2V_1$$

~~$V_1 = E_1$~~

$$\frac{E_1 - V_1}{R_3} + \frac{V_2 - V_1}{R_4} = 0 \Leftrightarrow \frac{E_1 - V_1}{R} + \frac{V_2 - V_1}{R} = 0$$

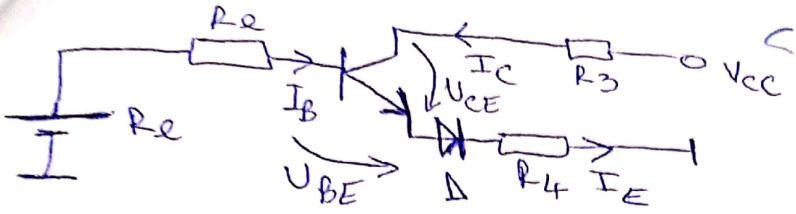
$$\Leftrightarrow E_1 - E_2 + V_2 = 0 \Rightarrow V_2 = E_2 - E_1$$

$$\frac{V_1 - V_2}{R_4} + \frac{0 - V_2}{R_5} + \frac{V_0 - V_2}{R_6} = 0 \Rightarrow V_0 = 3V_2 - V_1 = 3E_2 - 3E_1 - \frac{E_2}{2} = \frac{5E_2}{2} - 3E_1$$

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DEEA
Curs 3

Semiconductori



$$E_e = \frac{R_2}{R_1+R_2} V_{CC} = 5\,V$$

$$R_e = R_1 \parallel R_2 = 5K$$

$$E_e = I_B R_E + U_{BE} + U_D + I_E R_4$$

$$I_E \overset{N}{=} I_C = \beta I_B \Rightarrow I_B = \frac{I_E}{\beta}$$

$$I_E = \frac{E_e - U_{BE} - U_D}{\frac{R_e}{\beta} + R_4} = \frac{3,8}{0,05+3,8} \overset{N}{=} 1\,mA$$

$$V_{CC} - 0 = I_C R_3 + U_{CE} + U_D + I_E R_4$$

$$\Rightarrow U_{CE} = V_{CC} - I_C R_3 - U_D - I_E R_4 = 3,6\,V$$

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